CRISPR-CAS9 GENE EDITING — GRADUATE EXAM

Total Points: None | Questions: 5 | Date: April 17, 2026

Diagram-based questions on CRISPR-Cas9 Gene Editing

Question 1

10.0 points

Analyze the diagram below illustrating the Cas9 protein-sgRNA complex docking onto the target DNA sequence. Explain the roles of different domains of the Cas9 protein in binding with the target DNA.

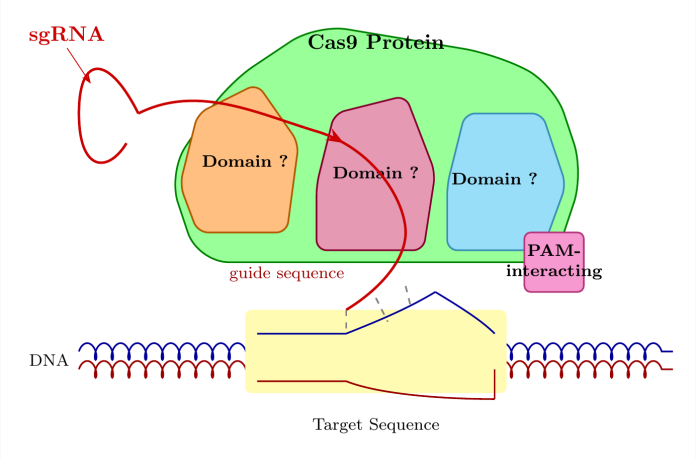

Fig. 1. Diagram illustrating the Cas9 protein-sgRNA complex docking onto the target DNA sequence. The diagram should show the Cas9 protein with labeled domains such as the RuvC, HNH, and PAM-interacting domains. The sgRNA should be shown bound to the Cas9 protein, guiding it to the target DNA. The DNA should be depicted as a double helix with the target sequence highlighted. Labels should clearly identify the Cas9 protein, sgRNA, and target DNA, as well as the specific domains of the Cas9 protein involved in binding. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • REC • HNH • RuvC The diagram will FAIL review if any of these appear.

Question 2

10.0 points

Interpret the diagram below showing the PAM sequence recognition and DNA unwinding mechanism of Cas9. How does the PAM sequence influence the specificity of DNA cleavage by Cas9?

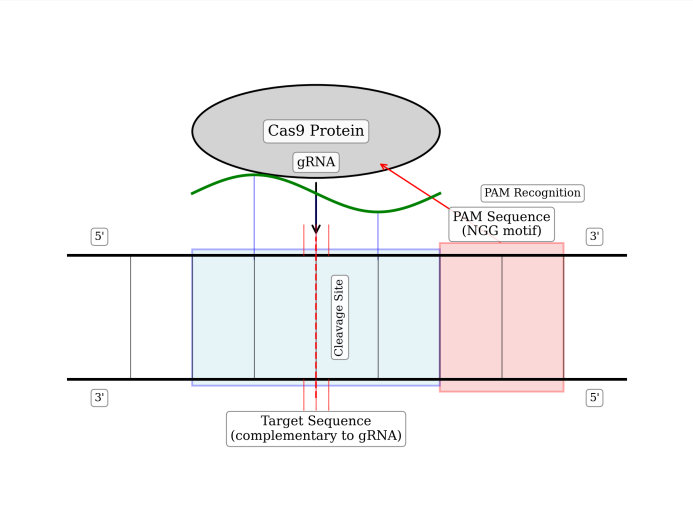

Fig. 2. Diagram showing the PAM sequence recognition and DNA unwinding mechanism of Cas9. Include the Cas9 protein, guide RNA, target DNA strand, and PAM sequence. Highlight the interaction between the PAM sequence and Cas9, showing how the DNA is unwound at the target site. Label the Cas9 protein, guide RNA, target DNA, and PAM sequence clearly. FORBIDDEN LABELS (do NOT render as text, legend entries, annotations, note boxes, or titles — these reveal the answer): • PI The diagram will FAIL review if any of these appear.

Question 3

10.0 points

Examine the diagram below illustrating the DNA double-strand break repair pathways: Non-Homologous End Joining (NHEJ) vs Homology-Directed Repair (HDR). Discuss the advantages and disadvantages of each pathway in gene editing applications.

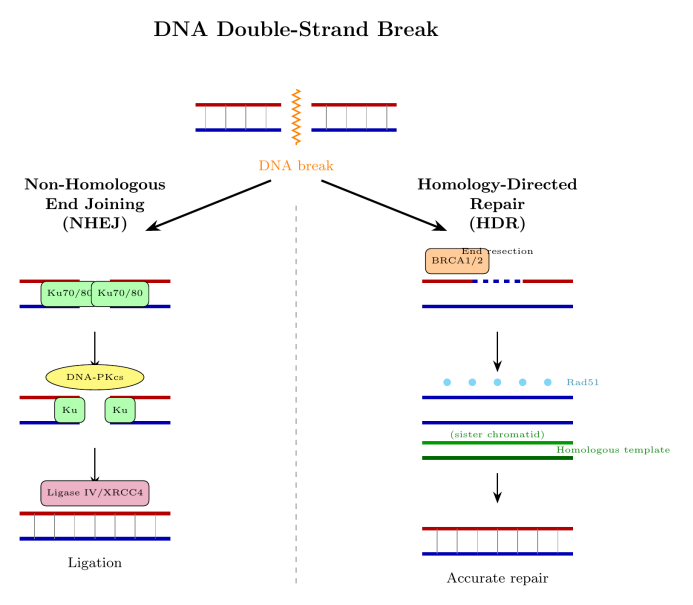

Fig. 3. Diagram illustrating the DNA double-strand break repair pathways: Non-Homologous End Joining (NHEJ) and Homology-Directed Repair (HDR). The diagram should show a DNA double helix with a break, with two pathways diverging from the break site. The NHEJ pathway should depict direct ligation of the broken ends, possibly with small insertions or deletions. The HDR pathway should show the use of a homologous template for accurate repair, highlighting the use of a sister chromatid or homologous chromosome. Include labels for key components like 'DNA break', 'NHEJ', 'HDR', 'homologous template', and 'ligation'.

Question 4

10.0 points

Evaluate the diagram below showing the base editing window of cytosine base editors (CBEs). What factors contribute to the editing window, and how does this influence precision in gene editing?

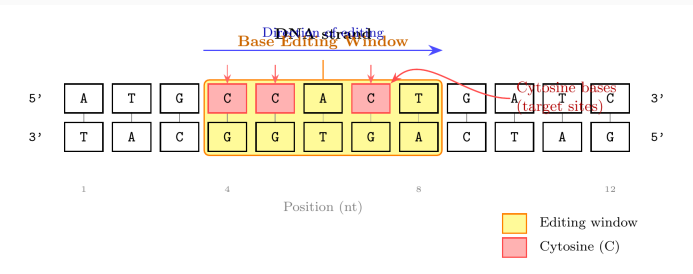

Fig. 4. MISMATCH 1: diagram shows

Question 5

10.0 points

Analyze the mismatch tolerance diagram shown below for predicting off-target effects in CRISPR-Cas9 gene editing. What are the implications of mismatch tolerance on off-target prediction and how can it be minimized?

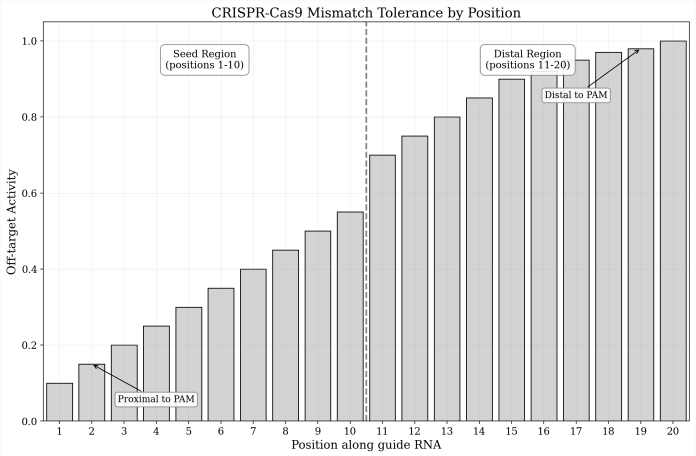

Fig. 5. CUMULATIVE LEAK CORRECTIONS (all must be fixed): Known answer terms to avoid: DNA, PAM; The diagram contains the label "PAM" in the annotations "Proximal to PAM" and "Distal to PAM", which is explicitly listed as a forbidden label for this question.. A bar chart showing CRISPR-Cas9 mismatch tolerance. The horizontal axis is labeled 'Position along guide RNA' and shows values from 1 to 20. The vertical axis is labeled 'Off-target Activity' and ranges from 0.0 to 1.0. The bars should show a generally increasing trend in height from left to right, indicating lower activity for positions 1-10 and higher activity for positions 11-20. There should be a label 'Seed Region (positions 1-10)' indicating the first ten bars. There should be a label 'Distal Region (positions 11-20)' indicating the last ten bars. A dashed vertical line may